

Crowd Counting Using Video Analytics



AI-powered Real-time Detection, Tracking, and Analytics

Tools Used:

Streamlit, YOLOv8, Plotly Express,
SQLite, Python.

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Training End Date :



Presented By:

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Agenda

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Introduction

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✦ This project addresses the growing need for efficient crowd management in public spaces such as malls, airports, and event venues by providing an automated solution for real-time people counting using video analytics. Leveraging advanced computer vision technologies like YOLOv8 for object detection and OpenCV for video processing, the system accurately detects, tracks, and counts individuals from live camera feeds. Built using Streamlit, the interactive dashboard not only visualizes real-time analytics and heatmaps but also enables zone-based monitoring, automated alerts, and easy data export. This approach overcomes the limitations of manual counting, offering a reliable, scalable, and user-friendly method to enhance safety, optimize operations, and make informed decisions in dynamic environments.

Problem Statement

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This project seeks to address this gap by automating the process of people counting through live camera feeds, using cutting-edge computer vision and deep learning techniques.

1. Automated real-time detection and counting of people from live camera feeds and uploaded videos/images using deep learning.
2. Interactive dashboard with live analytics, customizable zone monitoring, and instant alerts for overcrowding or unusual activity.
3. Easy export of detailed reports and visualizations, enabling administrators to make informed decisions for safety and efficiency.

The solution is designed to reliably detect and count individuals in video streams, deliver live analytics via an interactive dashboard, and support monitoring across multiple zones, all with high performance and real-time responsiveness.

Objective

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✦ In many crowded places like shopping malls, airports, railway stations, offices, or events, it's often difficult to keep track of how many people are present in a specific area. Usually, this is done manually by security staff or through guesswork, which can be slow and inaccurate. For example, during busy hours in a mall, it's hard to know if a particular zone like the food court or entrance area is becoming overcrowded. This can lead to safety problems, long waiting lines, or poor management.

- To automate people counting in public spaces using a live camera feed by leveraging computer vision and deep learning. ✦
- To provide real-time detection and accurate counting of people in video streams for improved security, analytics, and operational efficiency.
- To deliver live analytics, including density heatmaps and zone-wise statistics, through an interactive dashboard.
- To enable monitoring and alerting for multiple, user-defined zones with real-time performance.
- To generate exportable reports and provide ✦ an admin panel for managing cameras, reviewing footage, and monitoring logs.

Architecture Diagram



Tools And Technologies

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- Streamlit – Framework for building interactive dashboards and web UI.
- YOLOv8 (ultralytics) – Deep learning model for real-time person detection.
- OpenCV (cv2) – Library for video and image processing.
- Plotly Express – Visualization library for charts and analytics.
- NumPy and Pandas data handling and analytics
- SQLite – Lightweight database for storing user, camera, zone, and count data.
- Pillow (PIL) – Used for image processing
- JWT – Token based authentication and authorization.
- streamlit-drawable-canvas – Canvas interface for drawing and editing monitoring zones

Implementation Details

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This project can be implemented in a variety of real-world public and private spaces where crowd management, safety, and operational efficiency are a concern. For example, shopping malls and retail complexes can use this system to monitor peak crowd zones, triggering alerts when specific areas like food courts or entrances become overcrowded, thereby improving customer flow and reducing safety risks. Airports, railway stations, and other transport hubs can deploy the project for real-time monitoring of waiting halls, platforms, and check-in counters, ensuring that resources are allocated dynamically and helping prevent congestion during busy hours.

In event venues—such as stadiums, auditoriums, and exhibition centers—the system facilitates dynamic tracking of visitor numbers in different sections, aiding organizers in managing crowd surges efficiently, optimizing emergency responses, and adhering to occupancy regulations. Large office campuses, hospitals, and government buildings can leverage the solution to track daily occupancy trends, maintain social distancing in common areas, and streamline logistics during special events. Additionally, the solution finds utility in educational institutions, amusement parks, outdoor festivals, and anywhere crowd density plays a critical role in planning, safety, and user experience.

Detection & Tracking

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- The project uses YOLOv8 for detecting people in video frames, offering high accuracy and real-time performance on live or uploaded images and videos.
- Object tracking is accomplished with DeepSORT, which assigns unique IDs to every detected individual and tracks their movements across frames, displaying visual markers for each.
- Zone-based analytics allow users or admins to define areas of interest in camera feeds, monitor occupancy and detect overcrowding with real-time alerts.
- ✦ • An interactive dashboard built with Streamlit visualizes people counts, density heatmaps, and zone-wise statistics for better crowd management.
- The system supports role-based access, where users and admins see different controls and reports; authentication is secured with JWT tokens.
- Reporting and export functionality lets authorized users generate and download analytics summaries and logs in CSV or PDF format for further analysis or compliance.
- Secure camera and zone management tools are provided in the admin panel, allowing easy configuration, annotation, and review. ✦

Dashboard & Exportable Reports

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Larana, Inc.

Live Crowd Dashboard	Real-time display of people count and zone analytics	- Bar & line charts: current & historical counts. - Zone heatmaps: visualize density. - Live alerts panel. - Interactive zone/camera selector. - Table view: zone, time, alert status.
Admin Panel	Manage cameras, zones, users, and global alerts	- Tabbed interface for settings & users. - Zone/camera config tools. - Alert configuration. - System & user logs.
PDF Reports (Analytics)	Generate person/activity summaries and daily analytics	- ReportLab integration. - Individual/time-based reports. - Download from dashboard.
CSV Reports (Alert/History)	Export complete alert logs and count history	- Timestamps, alert type, user/zone details. - Users download own data, admins export entire history.

Admin panel



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In the Admin panel, the following actions and management features are typically available:

- User Management: Add, edit, or remove user accounts and assign roles. Admins can reset passwords and audit user activity for better security.
- Zone & Camera Configuration: Create, edit, or delete camera feeds and define monitoring zones. Customize alert escalation and notification channels for faster response.
- Alert Settings: Set and modify overcrowding thresholds, notification rules, and system alerts.
- Dashboard Controls: Access full analytics, reports, and real-time monitoring across all zones.
- System Logs: View and export logs of alerts, user actions, and system events for audit and security.
- Reports Export: Generate and download detailed analytics, alert history, and usage reports in PDF or CSV.
- Settings Management: Update system preferences, security policies, and integration settings.

These capabilities help administrators configure, control, and secure the crowd monitoring system efficiently.

Outputs of Milestone-1

1.Login Page

Login Register

Login

Email
jyoshna99@gmail.com

Password

Press Enter to submit form

Role
user

Sign In

2.Register Page

Create Account

Full Name
BACCHU VENKATA JYOSHNA

Email
jyoshna78@gmail.com

Password

Role
user

Register

3.Dashboard Page

Deploy

Hello, BACCHU VENKATA JYOSHNA (user)

Go to

- ☒ Analytics
- ☐ People Count
- ☐ (Admin only)

CrowdCount

Secure login • Red/Normal zones • Live analytics and alerts

Logout

Users	Cameras	Zones	Count Records
13	3	8	3591

Outputs of Milestone-2

1.Upload image:

After uploading an image, the system uses YOLOv8 to detect people and instantly displays the total count of detected individuals.

People Count & Zone Management

Select Camera

(None)

Count People

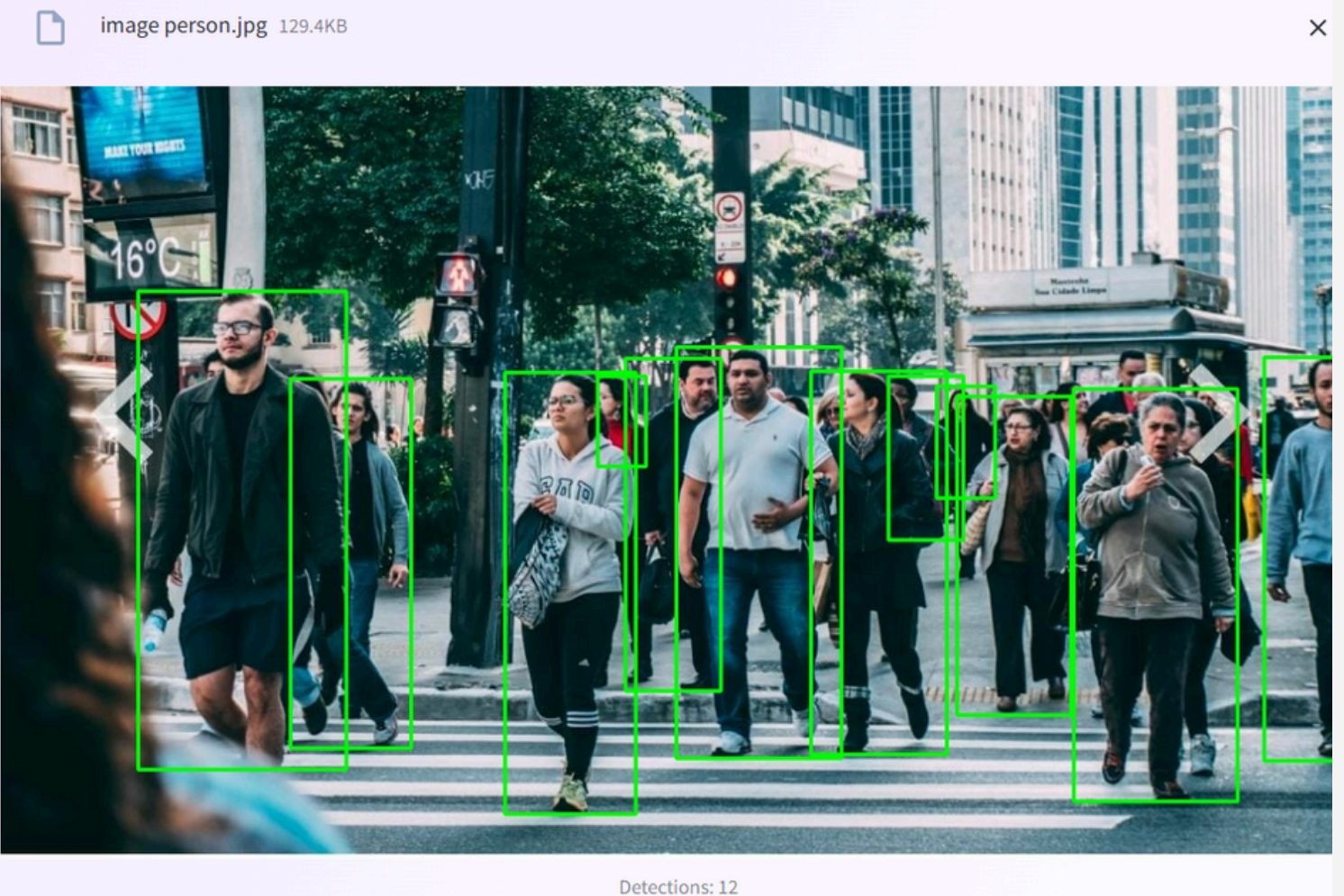
Image (draw zones) Video File Live (Webcam/IP)

Upload image



Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG



Outputs of Milestone-2

2.Upload Video:

People Count & Zone Management

Select Camera
(None) ▼

Count People

Image (draw zones) **Video File** Live (Webcam/IP)

Upload video

Drag and drop file here
Limit 200MB per file • MP4, AVI, MOV, MKV, MPEG4 Browse files

6387-191695740_large.mp4 7.2MB ×

Alert threshold
10 - +

Capture at second
1.00 - + Open Zone Editor (from Video)

Process with Analytics (zones/heatmap)



3.Live webcam:

People Count & Zone Management

Select Camera
(None)

Count People

Image (draw zones) Video File **Live (Webcam/IP)**

Connect to webcam or Paste URL

Source
0

☐ Start stream

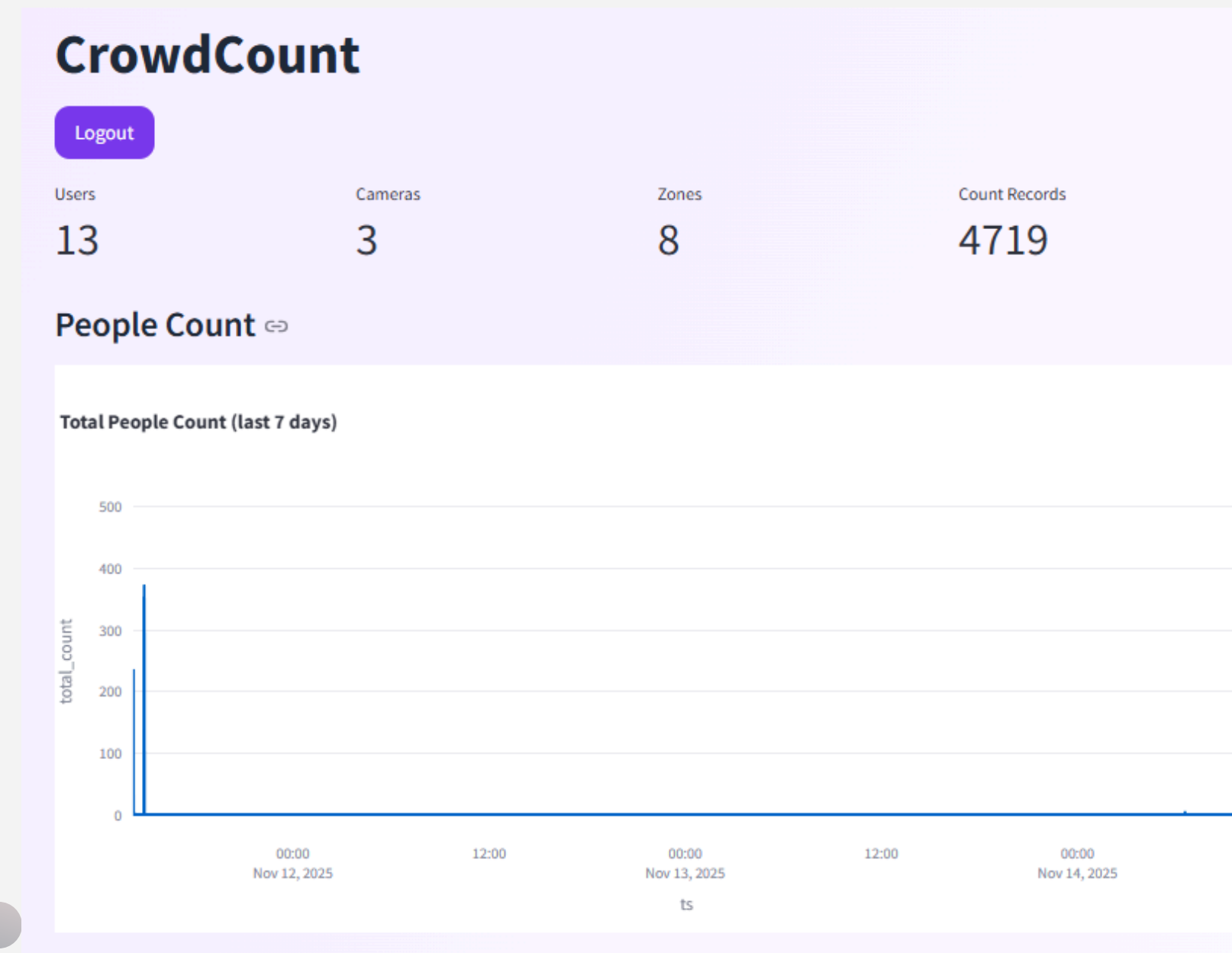
Live alert threshold
15

Snapshot & Open Zone Editor

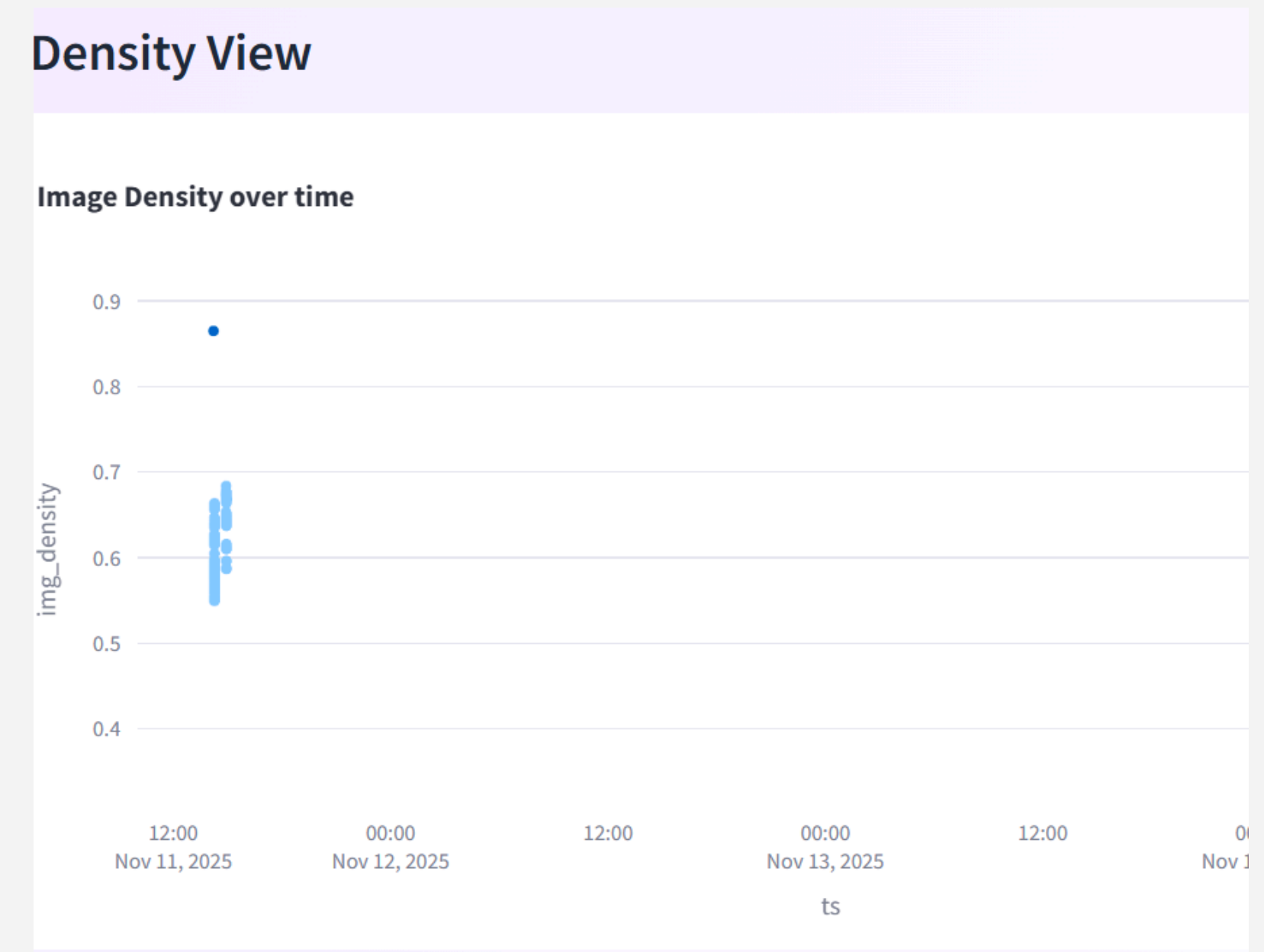
Outputs of Milestone-3

LIVE DASHBOARD: This dashboard gives users a real-time overview of people counting, image density, and zone occupancy based on camera feeds or sensors, with options to download the analyzed data.

1. People Count in Line chart



2. Density View in Scatter Plot



Outputs of Milestone-3

3. Zone Occupancy in BarChat:

CrowdCount provides real-time and historical zone occupancy analytics using stacked bar charts for smarter crowd monitoring



Outputs of Milestone-4

ADMIN PANEL:

- The admin panel in this project allows administrators to manage users, configure cameras and zones, set alert rules, view system logs, and generate detailed reports. It provides real-time dashboard controls and flexible settings management, ensuring all aspects of crowd monitoring are easy to control, secure, and customizable for any environment.

Create / Update User:

Create/Update User

Name

Email

Password

Role

user

Save User

Documentations & Reports

Hello, Bacchu Venkata Jyoshna (admin)

Go to

- Analytics
- People Count
- Admin

Admin Panel & Analytics

Users Cameras Logs Reports

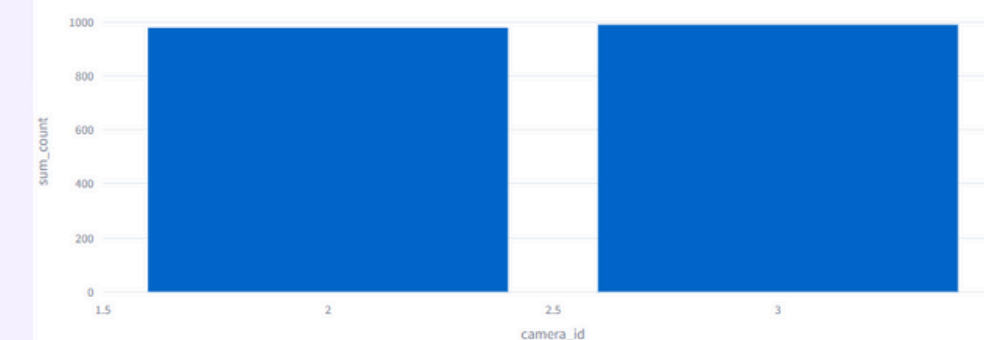
	id	name	email	role	created_at
0	14	Bacchu Venkata Jyoshna	jyoshna12@gmail.com	admin	2025-11-14T10:58:14.471979
1	13	BACCHU VENKATA JYOSHNA	jyoshna78@gmail.com	user	2025-11-14T09:00:40.827243
2	12	bvj	jyoshna3@gmail.com	user	2025-11-14T06:36:29.379982
3	11	BACCHU VENKATA JYOSHNA	jyoshna99@gmail.com	user	2025-11-11T12:24:44.157748
4	10	BACCHU VENKATA JYOSHNA	jyoshna12@gmail.com	user	2025-11-06T15:05:25.172674
5	9	BACCHU VENKATA JYOSHNA	jyoshna11@gmail.com	user	2025-11-06T14:59:26.201168
6	8	divya	jyoshna1@gmail.com	admin	2025-11-06T13:29:26.013303
7	7	BACCHU VENKATA JYOSHNA	jyoshna@gmail.com	user	2025-11-06T13:00:07.907369
8	6	Divya Sree	cdivyasree4@gmail.com	admin	2025-11-05T16:38:31.760764
9	5	C DIVYA SREE	cdivyasree48@gmail.com	user	2025-11-05T16:32:02.617188

Generate / Download Reports

Include last N days



Counts by Camera (Report)



Report saved: reports\report_20251114_110652.csv

Download This Report

Applications

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- **Banks or ticket counters:** Automatically monitor queue length and wait times, optimizing service counters allocation and customer experience.
- **Workplace safety monitoring:** Detect and track social distancing compliance, PPE (personal protective equipment) usage, or restricted area breaches in factories and construction sites.
- **Tourist attraction management:** Track visitor flow in museums, parks, or heritage sites to prevent bottlenecks and enhance security.
- **Emergency evacuation assistance:** In disaster scenarios, use real-time crowd tracking to guide safe and efficient evacuations.
- **Retail analytics for targeted promotions:** Analyze customer hotspots and dwell times to optimize product placement and promotional strategies.

Future Enhancements

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In the future, this project can be enhanced by adding multi-class detection (people, vehicles, objects), privacy features like face blurring, edge device support, real-time alerts via mobile or email, and animated heatmaps of people flow. Predictive analytics can help forecast crowd surges, and building a mobile app will enable remote monitoring. Integration with IoT devices for automated actions, customizable report options, and a scalable architecture for large venues will make the system even more powerful and flexible for various real-world applications.

✦ Conclusions

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This project not only improves crowd management and safety, but also supports data-driven decision making for administrators with real-time insights and detailed reports. The robust admin panel gives full control over cameras, zones, users, and alerts, making it easy to adapt the system to any public environment. With planned enhancements like multi-class detection, predictive analytics, and IoT integration, it is well-positioned to meet future requirements and handle a wide variety of complex scenarios efficiently. Overall, the solution is scalable, customizable, and provides the foundation for next-generation smart monitoring systems.



Thank You